

Brief A - Technical: Predictive Maintenance Approach

Audience: Maintenance / reliability engineer | Dataset: Mystery_Ops.csv | Focus: sensor logic, detection performance, and deployment limits

Purpose

The objective is to detect maintenance-related operating conditions early by combining recent sensor behavior with recent production performance. The analysis is centered on a sustained maintenance event on machine **NBI-P03**, where throughput fell sharply after the maintenance condition appeared.

Statistical validation: Welch's t-test found an average throughput reduction of **409.8 barrels** on NBI-P03. The 95% confidence interval for the reduction was **350.5 to 469.1 barrels**, with $p = 1.84 \times 10^{-12}$. This supports using recent throughput as a machine-health signal rather than relying on sensor readings alone.

Sensor and feature logic

Health feature	Engineering meaning
7-observation temperature mean	Recent thermal operating level
7-observation temperature-change RMS	Magnitude of recent temperature movement
7-observation temperature range	Short-term thermal instability
7-observation voltage-count standard deviation	Recent electrical variability
7-observation throughput mean	Recent production-performance level
7-observation throughput range	Short-term production instability

Leakage control: each rolling window is shifted by one observation before the current row is scored. The current maintenance label therefore does not build its own predictor history. **Target:** maintenance_flag = 1, missed_maintenance, or brief_equipment_fault. Operator blunders are excluded because they are human-operational events. **Model and validation:** standardized logistic regression with a chronological 75/25 split (2,140 training observations; 712 later holdout observations) and a 0.50 decision threshold.

Validation performance

99.44%	98.05%	99.82%	0.18%
Accuracy	Sensitivity	Specificity	False-positive rate

The chronological holdout contained 712 later observations. The classifier produced **151 true positives, 3 false negatives, 1 false positive**, and **557 true negatives**. Precision was **99.34%** and F1 was **98.69%**. For maintenance screening, the 98.05% sensitivity is the key result because missed degradation is operationally costly; the 0.18% false-positive rate indicates very little unnecessary inspection work in this sample.

Engineering interpretation and deployment recommendation

- Recent throughput history is the strongest signal. Temperature and voltage-count variability should be treated as supporting context rather than stand-alone trip criteria.
- Use the current result as a proof of concept, not a production acceptance test. Most positive labels come from one sustained NBI-P03 episode, which reduces failure-mode diversity.
- Before deployment, validate on additional machines and independent failure episodes, link alerts to confirmed work orders, and tune the alert threshold against the actual cost of missed failures versus unnecessary checks.

Brief B - Executive: Business Value of Early Maintenance Warning

Audience: Chief Financial Officer | Purpose: translate the pilot into cost, risk, and investment terms without technical detail

Business case

The approach watches recent machine readings and production output for signs that equipment performance is deteriorating. In the trial, it identified **151 of 154 maintenance cases**, or about **98 out of every 100**. It missed three cases and created only one unnecessary maintenance call across 558 normal checks. The business value is earlier intervention: less lost production, fewer emergency repairs, and fewer avoidable inspections.

Risk reduction: If similar performance holds on a wider set of machines, the company could materially reduce exposure to maintenance problems that continue unnoticed. The evidence is promising, but most of the maintenance examples came from one machine, so a broader pilot should come before a full-fleet financial commitment.

Illustrative annual savings and ROI

The dataset does not include repair, labor, or downtime costs. The figures below are therefore a transparent planning example, not observed company savings. They show how the pilot can be converted into a finance decision once site-specific costs are available.

Planning assumption	Illustrative amount
Avoidable maintenance disruptions per year	12
Average financial impact per disruption	\$6,000
Share of each disruption cost prevented by early action	60%
Trial success rate applied to savings estimate	98.0%
Annual cost to operate the warning approach	\$15,000

\$42,358	\$27,358	182%	4.2 mo.
Gross annual savings	Net annual savings	Estimated ROI	Simple payback

Decision recommendation

- Run a controlled wider pilot on more machines before making a full-scale spending decision.
- During the pilot, record the cost of downtime, emergency repairs, planned maintenance, and unnecessary maintenance visits so the savings estimate can be replaced with actual company figures.
- Approve wider rollout only if the measured reduction in disruption costs comfortably exceeds the annual cost of running the warning process.

Reflection: Switching between the two audiences was moderately challenging because the evidence did not change, but the useful language did. For an engineer, the explanation needs the sensor logic, recent-history window, missed cases, and unnecessary alerts. For a CFO, the same evidence must be translated into lost production, repair exposure, savings, and return on investment. The key skill was keeping the conclusion consistent while changing the level of detail and the decision focus.